



Review Article

A critical review of controlled clinical trials for peripheral neuropathic pain and complex regional pain syndromes

Wade S. Kingery^{a,b,*}

^aPhysical Medicine and Rehabilitation Service (117), Veterans Affairs Palo Alto Health Care System, 3801 Miranda Ave., Palo Alto, CA 94304, USA

^bDivision of Physical Medicine and Rehabilitation, Department of Functional Restoration, Stanford Medical School, Stanford, CA, USA

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Abstract

The purpose of this review was to identify and analyze the controlled clinical trial data for peripheral neuropathic pain (PNP) and complex regional pain syndromes (CRPS). A total of 72 articles were found, which included 92 controlled drug trials using 48 different treatments. The methods of these studies were critically reviewed and the results summarized and compared. The PNP trial literature gave consistent support (two or more trials) for the analgesic effectiveness of tricyclic antidepressants, intravenous and topical lidocaine, intravenous ketamine, carbamazepine and topical aspirin. There was limited support (one trial) for the analgesic effectiveness of oral, topical and epidural clonidine and for subcutaneous ketamine. The trial data were contradictory for mexiletine, phenytoin, topical capsaicin, oral non-steroidal anti-inflammatory medication, and intravenous morphine. Analysis of the trial methods indicated that mexiletine and intravenous morphine were probably effective analgesics for PNP, while non-steroidals were probably ineffective. Codeine, magnesium chloride, propranolol, lorazepam, and intravenous phentolamine all failed to provide analgesia in single trials. There were no long-term data supporting the analgesic effectiveness of any drug and the etiology of the neuropathy did not predict treatment outcome. Review of the controlled trial literature for CRPS identified several potential problems with current clinical practices. The trial data only gave consistent support for analgesia with corticosteroids, which had long-term effectiveness. There was limited support for the analgesic effectiveness of topical dimethylsulfoxide (DMSO), epidural clonidine and intravenous regional blocks (IVRBs) with bretylium and ketanserin. The trial data were contradictory for intranasal calcitonin and intravenous phentolamine and analysis of the trial methods indicated that both treatments were probably ineffective for most patients. There were consistent trial data indicating that guanethidine and reserpine IVRBs were ineffective, and limited trial data indicating that droperidol and atropine IVRBs were ineffective. No placebo controlled data were available to evaluated sympathetic ganglion blocks (SGBs) with local anesthetics, surgical sympathectomy, or physical therapy. Only the capsaicin trials presented data which allowed for meta-analysis. This meta-analysis demonstrated a significant capsaicin effect with a pooled odds ratio of 2.35 (95% confidence intervals 1.48, 3.22). The methods scores were higher ($P < 0.01$) for the PNP trials (66.2 ± 1.5 , $n = 66$) than the CRPS trials (57.6 ± 2.9 , $n = 26$). The CRPS trials tended to use less subjects and were less likely to use placebo controls, double-blinding, or perform statistical tests for differences in outcome measures between groups. There was almost no overlap in the controlled trial literature between treatments for PNP and CRPS, and treatments used in both conditions (intravenous phentolamine and epidural clonidine) had similar results. © 1997 International Association for the Study of Pain. Published by Elsevier Science B.V.

Keywords: Pain; Peripheral neuropathy; Complex regional pain syndromes; Analgesia; Clinical trials

1. Introduction

Peripheral neuropathic pain (PNP) is defined as pain initiated or caused by a primary lesion or dysfunction in

the peripheral nervous system (Merskey and Bogduk, 1994). This definition includes neuralgia and painful polyneuropathy. Neuralgia is pain in the distribution of a nerve or nerves. Painful polyneuropathies are usually symmetric and distal, affecting the feet and sometimes the hands. These conditions may present with or without paresthesia, hypoaesthesia, hyperalgesia and allodynia.

* Corresponding author. Tel.: +1 415 4935000, ext. 64768; fax: +1 415 8523470; e-mail: ma.kin@forysytthe.stanford.edu

The terms complex regional pain syndromes (CRPS) Type I and Type II are used to describe a syndrome of pain and sudomotor or vasomotor instability (Merskey and Bogduk, 1994). CRPS Type I (reflex sympathetic dystrophy) is defined as a syndrome that usually starts after a noxious event, is not limited to the distribution of a single peripheral nerve, and is disproportionate to the inciting event. The diagnosis requires: (a) pain, allodynia, or hyperalgesia disproportionate to injury; (b) evidence at some time of edema, changes in skin blood flow, or abnormal sudomotor activity in the region of pain; and (c) no other conditions that would otherwise account for the degree of pain and dysfunction. CRPS Type II (causalgia) is defined as a syndrome that starts after a nerve injury, and is not necessarily limited to the distribution of the injured nerve. The diagnostic criteria are the same as CRPS I. The difference between PNP and CRPS II is that the CRPS II diagnosis requires evidence of edema, cutaneous blood flow changes, or abnormal sudomotor activity.

The purpose of this review was to summarize the controlled trial data for PNP and CRPS, with a critical evaluation of the design and results of these studies. Specific objectives were to: (i) determine which diagnostic procedures and treatments were consistently supported by controlled trial data; (ii) evaluate studies which had conflicting results for the same treatment, to determine if there were large differences in the methods scores or other significant differences in the trial designs to explain the discrepancy in results; (iii) perform a meta-analysis of any treatment which was evaluated by three or more controlled trials with similar methodology and adequate data, but with contradictory results; (iv) identify potential flaws in design and differences in methods between PNP and CRPS trials; and (v) determine what treatments were used in both PNP and CRPS trials, and determine if there were treatment outcome differences specific to these diagnoses.

2. Methods

Recent reviews of neuropathic pain treatments referenced a total of 40 articles describing controlled trials (Fields, 1994; Calissi and Jaber, 1995; Galer, 1995a) and recent CRPS reviews cited a total of 17 controlled studies (Brody and Andary, 1993; Backonja, 1994; Subbarao and Blair, 1995; Haddox and Alstine, 1996). A combined manual and computer search of the literature found an additional 15 controlled studies, for a total of 72 articles published from 1969 to 1996. Reports were excluded that were only presented in abstract form, that presented data that was published in another article, and studies in which the results of PNP or CRPS patients could not be separated from patients with other disorders.

Each trial was evaluated for treatment effect, based on statistically significant differences between the treatment and the control groups at the 5% level. The primary out-

come measure or a combined measures score for the treatment effect was used whenever possible. Some trials failed to test for differences in treatment and control group outcomes, and some trials used multiple outcome measures and did not identify a primary outcome measure or evaluate a combined outcome score. The effectiveness of treatment for these trials was based on the authors' conclusion in the abstract or discussion. The global evaluation scale was used as the primary outcome measure for the capsaicin trials to allow for a meta-analysis. Some trials demonstrated no outcome differences in the treatment and control groups, but tested for differences in subsets of subjects selected by post hoc criteria. These trials were judged to have negative overall treatment effects, but positive subset effects were noted.

Each treatment that was compared to a control group was considered a trial. Many studies (12 PNP and 4 CRPS studies) used more than one treatment. Different methods of drug administration (e.g., oral, topical, intravenous, epidural, sympathetic ganglion injection, intravenous regional blocks) were considered different treatments, but different doses of the same drug were not considered different treatments.

All trials were scored according to the criteria listed in Appendix A. These criteria are a modified version of a methods assessment protocol that has been used in several critical review articles examining therapeutic interventions for low back pain (Koes et al., 1991a,b, 1994b, 1995; Koes and Hoogen, 1994a; Heijden et al., 1995). The scores for each methods criteria and the overall methods scores were compared for the PNP and CRPS trials. Student's *t*-test was used to determine differences, and differences were considered significant for $P < 0.05$. All data is presented as the mean $\pm$ standard error of the mean. The use of meta-analytic methods was considered whenever there were three or more trials with contradictory results. In order to be included in a meta-analysis the studies had to use similar treatment protocols, outcome measures, and present data on the number of subjects treated and showing improvement in both the treatment and control groups. The Mantel-Haenszel method (Mantel and Haenszel, 1959) was used to calculate the odds ratio for each trial (Yusuf et al., 1985).

3. Controlled treatment trials

3.1. Peripheral neuropathic pain

A total of 50 PNP studies tested 31 different treatments in 66 trials. Table 1 summarizes 47 articles describing placebo controlled treatment trials for PNP. Three additional articles were identified that only compared one antidepressant drug to another (Watson and Evans, 1985; Watson et al., 1992), or to acetylsalicylic acid (Langhor et al., 1982).

Tricyclic antidepressants provide moderate analgesia for thermally, mechanically and electrically evoked pain in normal subjects (Bromm et al., 1986; Coquoz et al., 1991;

Poulsen et al., 1995). Antidepressants also can produce partial pain relief in diabetic (Kvinesdal et al., 1984; Max et al., 1987, 1991, 1992; Sindrup et al., 1989, 1990a,c, 1992a,b), postherpetic (Watson et al., 1982, 1992; Watson and Evans, 1985; Max et al., 1988b; Kishore-Kumar et al., 1990), and peripheral nerve injury (Langhor et al., 1982) neuropathic pain. Both spontaneous and evoked (Watson et al., 1992) neuropathic pain is relieved, but the long-term analgesic efficacy has not been established. The plasma concentration probably correlates with the analgesic effect (Kvinesdal et al., 1984; Max et al., 1988b; Sindrup et al., 1990c) and the

pain relief appears to be independent of the antidepressant effect of these drugs (Watson et al., 1982; Max et al., 1987, 1992; Sindrup et al., 1990c).

All trials using tricyclic antidepressants (amitriptyline, desimpramine, imipramine and clomipramine) have demonstrated superiority to either placebo (Watson et al., 1982; Kvinesdal et al., 1984; Max et al., 1988b; Sindrup et al., 1989, 1990a,c, 1992b), placebo with a medication to mimic tricyclic side-effects (Max et al., 1987, 1991, 1992; Kishore-Kumar et al., 1990), non-tricyclic antidepressants (Watson and Evans, 1985; Sindrup et al., 1990a, 1992b; Max et al.,

Table 1

Placebo controlled drug trials in peripheral neuropathic pain

Drug	Route	Neuropathy	Analgesia	Trial (methods score)
Antidepressants	Oral	Diabetic	Yes	Max et al., 1987 (76), Max et al., 1991 (72), Max et al., 1992 (79), Kvinesdal et al., 1984 (61), Sindrup et al., 1989 (57), Sindrup et al., 1990a (69), Sindrup et al., 1990b (69), Sindrup et al., 1991 (51), Sindrup et al., 1992a (64), Sindrup et al., 1992b (68)
		Postherpetic	Yes	Kishore-Kumar et al., 1990 (72), Max et al., 1988b (79), Watson et al., 1982 (66)
Lidocaine	Intravenous	Diabetic	Yes	Kastrup et al., 1987 (68)
		Postherpetic	Yes	Rowbotham et al., 1991 (68)
		Nerve injuries	Yes	Marchettini et al., 1992 (40), Wallace et al., 1996 (65)
	Topical	Postherpetic	Yes	Rowbotham et al., 1995 (75), Rowbotham et al., 1996 (67)
Mexiletine	Oral	Diabetic	Yes	Dejgard et al., 1988 (79)
			No ^a	Stracke et al., 1992 (75)
		Nerve injuries	Yes	Chabal et al., 1992 (63)
Ketamine	Intravenous	Postherpetic	Yes	Eide et al., 1994 (71)
		Nerve injuries	Yes	Felsby et al., 1995 (65)
				Nikolajsen et al., 1996 (60)
	Subcutaneous	Postherpetic	Yes	Eide et al., 1995 (43)
Phenytoin	Oral	Diabetic	Yes	Chadda and Mathur, 1978 (63)
			No	Saudek et al., 1977 (71)
Carbamazepine	Oral	Diabetic	Yes	Wilton, 1974 (68)
			Yes	Rull et al., 1969 (54)
Capsaicin	Topical	Diabetic	Yes	Capsaicin Study Group, 1991 (76)
			No	Chad et al., 1990 (60)
		Polyneuropathy	No	Low et al., 1995 (92)
		Postherpetic	Yes	Bernstein et al., 1989 (71)
			No ^a	Watson et al., 1993 (82)
Clonidine	Topical	Diabetic	No ^a	Zeigler et al., 1992 (70)
			Yes	Byas-Smith et al., 1995 (73)
	Oral	Postherpetic	Yes	Max et al., 1988a (78)
	Epidural	Cancer- neuropathic	Yes	Eisenach et al., 1995 (58)
Non-steroidal anti-inflammatory	Oral	Diabetic	Yes	Cohen and Harris, 1987 (52)
		Radiculopathy	No	Weber et al., 1993 (76)
		Postherpetic	No	Max et al., 1988a (78)
	Topical	Postherpetic	Yes	Benedittis et al., 1992 (65), Benedittis and Lorenzetti, 1996 (70)
Morphine	Intravenous	Nerve injuries	Yes	Kupers et al., 1991 (60)
			No	Arner and Meyerson, 1988 (40)
		Postherpetic	Yes	Rowbotham et al., 1991 (68)
			No	Eide et al., 1994 (71)
Codeine	Oral	Postherpetic	No	Max et al., 1988a (78)
Magnesium	Intravenous	Nerve injuries	No	Felsby et al., 1995 (65)
Propranolol	Oral	Nerve injuries	No	Scadding et al., 1982 (56)
Lorazepam	Oral	Postherpetic	No	Max et al., 1988b (79)
Phentolamine	Intravenous	Polyneuropathy	No	Verdugo et al., 1994 (55)

^aSubset of patients had analgesia.

1992), lorazepam (Max et al., 1988b), or acetylsalicylic acid (Langhor et al., 1982). The relative analgesic efficacy of the various tricyclics is not established, although amitriptyline, desimpramine and clomipramine are probably equivalent (Max et al., 1992; Sindrup et al., 1990c), and amitriptyline is more effective than maprotiline (Watson et al., 1992). Some serotonin receptor reuptake inhibitors (paroxetine and citalopram) may also provide analgesia (Sindrup et al., 1990a, 1991, 1992a), although probably not as effectively as the tricyclics (Watson and Evans, 1985; Sindrup et al., 1990a, 1992b; Max et al., 1992).

The tricyclics are initiated with a low bedtime dose (10 or 25 mg) and gradually increased on a weekly basis up to 150 mg/day if analgesia is inadequate. Serum concentrations may help guide the clinician in adjusting dosages in patients with a poor response (Sindrup et al., 1990b), but in many patients (7–58%) tricyclic antidepressants either do not provide good analgesia or have side-effects which limit the effective dosage (Max, 1994). The mechanism of the tricyclic analgesia is unknown, but these drugs block norepinephrine and serotonin reuptake, block α 1-adrenergic receptors, reduce sympathetic efferent activity and block the hyperalgesia induced by intrathecal *N*-methyl-D-aspartate (Max, 1994; Eisenach and Gebhart, 1995).

Intravenous lidocaine has no effect on ischemic and thermally evoked pain in normal subjects at serum levels below 4 μ g/ml, but higher serum levels of lidocaine are probably analgesic (Rowlingson et al., 1980; Boas et al., 1982; Nielsen et al., 1991). Infusion of lidocaine has been used as a diagnostic test to identify patients that may potentially benefit from orally administered lidocaine analogues. Controlled studies in diabetic (Kastrup et al., 1987), postherpetic (Rowbotham et al., 1991), and peripheral nerve injury (Marchettini et al., 1992; Wallace et al., 1996) patients with PNP have demonstrated short-term analgesia with intravenous lidocaine. Allodynia can be relieved (Marchettini et al., 1992; Wallace et al., 1996) and analgesia is achieved at mean serum levels between 1.5 and 2.0 μ g/ml (Boas et al., 1982; Rowbotham et al., 1995; Wallace et al., 1996). After stopping the lidocaine infusion no residual analgesia is observed (Kastrup et al., 1987). Topical lidocaine applied over postherpetic neuropathic painful skin provides a modest short-term analgesia when compared to placebo or lidocaine application over a contralateral site (Rowbotham et al., 1995, 1996).

Mexiletine is an orally administered lidocaine analogue which has been tested in several PNP trials at dosages between 450 and 750 mg per day. The higher dosages have provided effective analgesia in small controlled studies of diabetic (Dejgard et al., 1988) and peripheral nerve injury (Chabal et al., 1992) patients. One large multicenter controlled trial just failed to reach significance for an analgesic effect with 225–675 mg per day in diabetic neuropathic pain (Stracke et al., 1992). Retrospective analysis indicated that a subset of the patients in this study with symptoms of stabbing or burning pain did have an analgesic

response, but this needs confirmation using prospective criteria.

Ketamine is a *N*-methyl-D-aspartic acid (NMDA) receptor antagonist which may inhibit hyperexcitability of spinal nociceptive neurons. Intravenous high dose ketamine (500 μ g/kg bolus, 560 μ g/kg/h infusion) raises pain thresholds to pressure and flexor reflex thresholds to repetitive electrical stimulation, while lowering the perceived pain intensity of a single noxious electrical stimulation (Arendt-Nielsen et al., 1995). There is no effect on heat pain thresholds or flexor reflex thresholds to a single noxious electrical stimulus. Controlled trials have used intravenous ketamine (100–200 μ g/kg bolus, 0–420 μ g/kg/h infusion) in postherpetic (Eide et al., 1994) and peripheral nerve injury (Felsby et al., 1995; Nikolajsen et al., 1996) PNP patients. These trials have demonstrated ketamine analgesia for spontaneous pain and mechanical pain thresholds, mechanical allodynia and mechanical summated repetitive stimulation, but no ketamine effect on heat or cold pain thresholds or heat summated repetitive stimulation. Continuous subcutaneous infusion of ketamine has also been shown to reduce spontaneous pain in postherpetic PNP in direct relationship to the serum concentrations of ketamine and nor-ketamine (Eide et al., 1995). Unfortunately, the continuous infusion of ketamine is not feasible due to intolerable side-effects including painful indurations at the infusion site and psychomimetic effects such as sedation, slowed reaction times, and hallucinations (Arendt-Nielsen et al., 1995; Eide et al., 1995). Using intravenous ketamine as a diagnostic test for identifying patients that respond to NMDA antagonists has no clinical application, since there is no controlled trial data on the use of other NMDA antagonists in PNP or CRPS.

Anticonvulsant analgesia in diabetic PNP has been examined in four randomized, crossover design, double-blinded trials. None of studies adjusted the anticonvulsant dosage to maintain target serum levels. The phenytoin (300 mg/day) results are mixed, with one trial reporting effective analgesia after 2 weeks (Chadda and Mathur, 1978) and another investigation finding no analgesic effect over 23 weeks (Saudek et al., 1977). Carbamazepine (600 mg/day) had an analgesic effect in one arm and not in the other arm of a crossover study at 7 days, but there was no combined data analysis for all subjects (Wilton, 1974). Another study found a greater than 50% relief on the investigator global evaluation scale in 66% of carbamazepine (600 mg/day) treated patients and 20% of placebo treated patients after 2 weeks treatment, but no statistical analysis was performed (Rull et al., 1969).

Capsaicin, the pungent component of hot peppers, can enhance the release and inhibit the reuptake of substance P and other neuropeptides from terminals of unmyelinated polymodal afferents. Topical application of capsaicin in normal subjects has no effect on touch, coolness, warmth, mechanical pain, or cold pain thresholds. There is a brief period of heat hyperalgesia and a burning sensation immediately after capsaicin application, but then heat pain thresh-

olds return to normal. After 4 weeks of applying topical capsaicin (with either 0.075% or 0.75% concentrations) there is a decrease in the magnitude estimation of supra-threshold heat pain in normal subjects (Simone and Ochoa, 1991; Beydoun et al., 1996).

The analgesic properties of topical 0.075% capsaicin have been investigated in three polyneuropathy (Chad et al., 1990; Capsaicin Study Group, 1991; Low et al., 1995) and two postherpetic (Bernstein et al., 1989; Watson et al., 1993) PNP studies. The investigator global evaluation for symptom relief at the end of the treatment (4–8 weeks) indicated capsaicin was better than placebo in two trials (Bernstein et al., 1989; Capsaicin Study Group, 1991), with two other trials showing a trend for greater improvement in the capsaicin groups (Chad et al., 1990; Watson et al., 1993). Some patients complained of burning and/or erythema after topical application of capsaicin, which may prevent continued use and makes true patient blinding in these trials problematic. The study with the largest placebo effect (67%) was the only study to use a placebo which caused skin erythema and stinging, and this study did not demonstrate a trend in favor of capsaicin (Low et al., 1995). A meta-analysis of these five studies demonstrated improved global evaluation scores in 167/247 (68%) of capsaicin treated and 122/257 (47%) of placebo treated patients. A significant capsaicin effect was demonstrated with a pooled odds ratio for all five trials of 2.35 (95% confidence intervals 1.48, 3.22).

Two trials tested transdermal clonidine patches (0.3 mg/day) applied over painful skin in diabetic PNP patients. Both studies found no clonidine analgesia over 3–6 weeks of continuous treatment, but a subset of the total patient population, identified by retrospective (Zeigler et al., 1992) and prospective (Byas-Smith et al., 1995) criteria did have analgesia. Another trial found that a single oral dose of clonidine (0.2 mg) provided analgesia in postherpetic neuralgia patients (Max et al., 1988a). Continuous infusion epidural clonidine (30 μ g/h) for 14 days had no effect on patient controlled morphine analgesia requirements, but did significantly reduce the reported pain intensity in a study of cancer patients with neuropathic pain (Eisenach et al., 1995).

Non-steroidal anti-inflammatory drugs have had mixed results in several neuropathic pain trials. The study with the lowest methods score found ibuprofen (2400 mg/day) or sulindac (400 mg/day) were both effective analgesics for diabetic neuropathic pain (Cohen and Harris, 1987). A 4-week course of piroxicam (20 mg/day) was ineffective in a large double-blind, parallel, randomized trial of lumbosacral radiculopathy patients (Weber et al., 1993). A single dose of ibuprofen (800 mg) had no analgesic effect in a large crossover trial of postherpetic neuralgia patients (Max et al., 1988a). A single dose of a topical aspirin/diethyl ether mixture provided mild analgesia in postherpetic neuralgia patients (Benedittis et al., 1992; Benedittis and Lorenzetti, 1996).

Intravenous morphine produces analgesia for evoked pain in normal subjects, both in a dose and a plasma concentration dependent manner (Price et al., 1985; Hill et al., 1990). Several controlled trials have examined the analgesic efficacy of opioids in chronic neuropathic pain patients. No significant analgesic effect was seen in postherpetic neuralgia patients following a single 120-mg dose of codeine (Max et al., 1988a). A study with a low methods score tested eight neuropathic pain patients who had failed prior opioid therapy and were scheduled for intracerebral or dorsal column stimulators (Arner and Meyerson, 1988). Morphine (15 mg i.v. bolus) provided partial relief in only one of these patients. One study found that a low dose of morphine (75 μ g/kg i.v. bolus) in postherpetic PNP patients failed to relieve spontaneous pain, but did reduce mechanical allodynia (Eide et al., 1994).

Two other placebo controlled trials with average methods scores demonstrated analgesia with higher morphine doses (300 μ g/kg i.v. bolus) in PNP patients (Kupers et al., 1991; Rowbotham et al., 1991). Opioid analgesia has been correlated to dose and serum drug levels in neuropathic patients (Rowbotham et al., 1991; Cherny et al., 1994). Evoked pain can also be transiently relieved with morphine in most patients with postherpetic neuralgia (Rowbotham et al., 1991). There is a wide range of analgesic responses to opioids in both neuropathic and nociceptive pain patients, and some data suggest that neuropathic pain is less opioid responsive than nociceptive pain (Cherny et al., 1994). The only study to look at serum morphine levels and analgesia in neuropathic pain demonstrated significant analgesia with a mean peak serum level of 68 ng/ml (Rowbotham et al., 1991), which is less than the EC_{50} for analgesia seen in normal subjects for electrically evoked dental pain (Hill et al., 1990).

Watson has recommended guidelines for the use of opioids in postherpetic neuralgia (Watson, 1995). The use of opioids in neuropathic pain should be a last resort. Oxycodone 5–10 mg every 4 h is a practical therapeutic approach. Patients with a history of chemical dependency should be excluded and there should be one prescriber and one dispenser. Patients need to be seen regularly to monitor pain and record drug utilization.

Several other drugs were ineffective in PNP patients. Magnesium chloride (0.16 mmol/kg i.v. bolus, 0.16 mmol/kg per h infusion) was ineffective in postherpetic neuralgia (Eide et al., 1994). A 2-week course of propranolol (a beta-adrenergic receptor antagonist, 240 mg/day) did not provide relief in peripheral nerve injury patients (Scadding et al., 1982). Lorazepam (a benzodiazepine, 0.5–6 mg/day) for 6 weeks had no analgesic properties in postherpetic neuralgia patients (Max et al., 1988b). Intravenous systemic phentolamine (an alpha-adrenergic receptor antagonist, 35 mg i.v. bolus) was an ineffective analgesic for polyneuropathy patients with spontaneous and evoked neuropathic pain (Verdugo et al., 1994).

Table 2

Placebo controlled drug trials in complex regional pain syndromes

Drug	Route	Analgesia	Trial (methods score)
Guanethidine	Intravenous regional block	Yes	Glynn et al., 1981 (54)
		No	Rocco et al., 1989 (68), Blanchard et al., 1990 (71), Jadad et al., 1995 (68), Ramamurthy et al., 1995 (83)
Reserpine	Intravenous regional block	No	Rocco et al., 1989 (68), Blanchard et al., 1990 (71)
Droperidol	Intravenous regional block	No	Kettler and Abram, 1988 (44)
Atropine	Intravenous regional block	No	Glynn et al., 1993 (72)
Bretylum	Intravenous regional block	Yes	Hord et al., 1992 (57)
Ketanserin	Intravenous regional block	Yes	Hanna and Peat, 1989 (48)
	Intravenous	No	Bounameaux et al., 1984 (45)
Corticosteroids	Oral	Yes	Christensen et al., 1982 (71), Braus et al., 1994 (51)
Calcitonin	Intranasal	Yes	Gobelet et al., 1992 (80)
		No	Bickerstaff and Kanis, 1991 (82)
Clonidine	Epidural	Yes	Rauck et al., 1993 (59)
Phentolamine	Intravenous	Yes	Raja et al., 1991 (41)
		No	Verdugo and Ochoa, 1994 (52)

3.2. Complex regional pain syndromes

A total of 22 articles for CRPS were identified, and included 26 controlled trials testing 17 different treatments. Table 2 summarizes 17 articles describing placebo controlled trials which investigated 10 different treatments for CRPS. Five other articles compared two treatments to each other (Bonelli et al., 1983; Glynn and Casale, 1993; Geertzen et al., 1994), or one treatment to no treatment (Gobelet et al., 1986; Field and Atkins, 1993).

Intravenous regional blocks (IVRBs) with guanethidine will deplete noradrenaline in the post-ganglionic axon. Table 3 illustrates that guanethidine IVRBs in CRPS patients are ineffective analgesics compared to placebo or no treatment (Rocco et al., 1989; Blanchard et al., 1990; Field and Atkins, 1993; Ramamurthy et al., 1995). One study found a decrease in spontaneous pain relative to baseline levels at 1 h after a guanethidine IVRB, but a later study by the same investigator failed to demonstrate any guanethidine effect (Glynn et al., 1981; Jadad et al., 1995). One trial demonstrated a better recovery in early CRPS after 3 weeks of topical dimethylsulfoxide (DMSO) than after six guanethidine blocks (Geertzen et al., 1994).

One potential criticism of the guanethidine data is that four studies used only one IVRB, but multiple blocks are frequently used for the treatment of CRPS. The study with the highest methods score found no difference over 26 weeks between patients receiving one, two or four guanethidine blocks, demonstrating that multiple blocks are just as ineffective as a single IVRB (Ramamurthy et al., 1995).

Numerous other drugs have been used in controlled IVRB trials (Table 4). Bretylum (which depletes noradrenaline in the post-ganglionic axon) provided significantly longer analgesia (20 ± 18 days) than lidocaine (2.7 ± 4 days) (Hord et al., 1992). Analgesia has also been reported for 3 weeks following several ketanserin blocks (a serotonin type 2 receptor antagonist) (Hanna and Peat, 1989). These

two trials, with a combined total of 16 patients, need to be confirmed with larger investigations. Two trials with reserpine (which depletes noradrenaline in the post-ganglionic axon) did not demonstrate any analgesia (Rocco et al., 1989; Blanchard et al., 1990). Atropine (a muscarinic cholinergic antagonist) and droperidol (an alpha-adrenergic antagonist) had no analgesic effects following IVRBs in chronic CRPS (Kettler and Abram, 1988; Glynn et al., 1993).

Table 5 lists eight more controlled drug trials for CRPS. Corticosteroids were effective analgesics in several trials with early CRPS patients (Christensen et al., 1982; Braus et al., 1994). Christensen et al. found corticosteroids effective 12 weeks after initiation of treatment.

Calcitonin administered subcutaneously or by intranasal spray over 3–4 weeks had mixed results in early CRPS, with two studies finding no difference between calcitonin and control (Gobelet et al., 1986; Bickerstaff and Kanis, 1991), and one study showing a benefit after calcitonin treatments (Gobelet et al., 1992). The two nasal calcitonin studies had contradictory results, despite almost identical designs, patient populations, and high methods scores (Bickerstaff and Kanis, 1991; Gobelet et al., 1992). The primary difference between these two studies was the reliability of the different outcome measures. The study that demonstrated a calcitonin analgesic effect used a 4-point ordinal scale for subjective measurement of tenderness, range of motion and edema (Gobelet et al., 1992). The study finding no calcitonin effect (Bickerstaff and Kanis, 1991) used quantitative outcome measures for dolorimetry, grip strength, and hand volume measurements which were readily reproducible (coefficient of variation range was 3.8–8.3%). The use of reliable continuous objective outcome data gives additional credence to the Bickerstaff and Kanis study.

Epidural clonidine (an alpha 2-adrenergic receptor agonist) had an analgesic effect in chronic CRPS, but

Table 3
Controlled trials of intravenous regional guanethidine blocks in CRPS

Trial (methods score)	Subjects and design	Diagnosis (duration of symptoms)	Treatment (guanethidine)	Control treatment (tourniquet time)	Outcome measure (follow up)	Results: guanethidine vs. control	Results: control
Glynn et al., 1981 (54)	n = 28; C, SB	+SGB, 4/7 symptoms and signs of CRPS (?)	G: 10 mg in 10 ml saline, 1 block	Saline: 10 ml, 1 block (10 min)	VAS (I) (1 h)	Significant at 1 h (NS at 1 week from baseline)	NS at 1 h
Bonelli et al., 1983 (49)	n = 9; P, O, R	4/6 signs of CRPS; G: (18 weeks); SGB: (28 weeks)	G: 20 mg in 25 ml saline, 4 blocks over 16 days	SGB: B 15 ml, 0.5%, 8 blocks over 16 days	VAS (I) (16 days)	NS at all times	Significant at 1 h, persists to 16 days
Rocco et al., 1989 (68)	n = 12; C, DB, R	+SGB, allodynia, pain, vasomotor (?)	G: 20 mg in 50 ml 0.5% L, 1 block	L: 50 ml, 0.5%; (20 min)	VAS (I) (90 min); NPS (1 week)	NS at all times	Average VAS ↓ 72% at 90 min
Blanchard et al., 1990 (71)	n = 12; C, DB, R	+ SGB, signs and symptoms (2.9 years)	G: 20 mg UE, 30 mg LE, in 30–50 ml saline, 1 block	Saline: 30–50 ml (20 min)	↓ VAS (I) > 50% = relief (until no relief)	NS at all times	83% with relief at 30 min
Field and Atkins, 1993 (46)	n = 10; P, O	Colle's Fx, ↓ range vasomotor, edema, dolorimetry (12 weeks)	G: ? mg? ml, 2–4 blocks and physical therapy	Physical therapy	Five physical measures (24 weeks)	NS except for dolorimetry at 20 and 24 weeks	?
Geertzen et al., 1994 (59)	n = 13; P, O, R	3/6 symptoms and signs of CRPS (3–12 weeks)	G: ? mg? ml, 6 blocks over 3 weeks	DMSO (50% in water) topical 4 × day for 3 weeks	VAS (I); three clinical scales, ADLs (9 weeks)	Significantly worse than DMSO at 7 and 9 weeks	Significant, combined score was better than G at 7 and 9 weeks
Jadad et al., 1995 (68)	n = 8; C, DB, R	+G block, 4/7 symptoms and signs of CRPS (?)	G: 10 and 30 mg UE, 20 and 30 mg LE in 25–50 ml saline, 1 blockdose	Saline: 25 ml UE, 50 ml LE (15 min)	VAS (I) (R); (1 week, then until pain at baseline)	NS at all times	Average VAS ↓ 37% over 1 h, 14% over 1 week
Ramamurthy et al., 1995 (83)	n = 20; P, DB, R	Allodynia, hyperalgesia (6 weeks)	G: 20 mg UE, 40 mg LE, 30–75 ml 0.5% L, 1, 2, or 4 blocks over 12 days	L: 30–50 ml UE, 40–75 ml LE, 0.5%, 0–3 blocks (20 min)	MPQ and global evaluation (4 days) (26 weeks)	NS at 4 days (no dose response differences for G to 26 weeks)	Average MPQ ↓ 27% after 4 days

C, crossover; B, bupivacaine; DB, double-blind; Fx, fracture; G, guanethidine; (I), intensity; L, lidocaine; LE, lower extremity; MPQ, McGill Pain Questionnaire; n, number of subjects in smallest group after randomization; NPS, numeric pain scale; NS, not significant; O, open; P, parallel; q, each; QST, quantitative sensory testing; R, randomized; (R), relief; SB, single-blind; SGB, sympathetic ganglion block; UE, upper extremity; VAS, visual analogue scale for pain.

Table 4
Controlled trials of intravenous regional drug blocks in CRPS

Trial (methods score)	Subjects and design	Diagnosis (duration of symptoms)	Treatment drug	Control treatment (tourniquet time)	Outcome measure (follow-up)	Results: drug vs. control (R)	Results: control
Kettler and Abram, 1988 (44)	n = 6; C, DB, R	+SGB, clinical signs CRPS (3.1 years)	Droperidol 2.5 mg in 30–50 ml saline, 1 block	Saline 30 ml UE, 50 ml LE (15 min)	VAS (I) (2 weeks)	1/5 with ↓ VAS > 40%	3/4 with ↓ VAS > 40%
Hanna and Peat, 1989 (48)	n = 9; C, DB, R	Pain, vasospasm, edema, atrophic (2.1 years)	Ketanserin 10 mg, 10 mg UE, 20 mg LE, 30 ml saline, 2 blocks	Saline 30 ml, 2 blocks (15 min)	VAS (I) (2 weeks)	Significant at 1 and 2 weeks from baseline	NS at 1 week
Rocco et al., 1989 (68)	n = 12; C, DB, R	+SGB, allodynia, pain, vasomotor (?)	Reserpine 1.5 mg in 50 ml 0.5% L, 1 block	Lidocaine 0.5% 50 ml (20 min)	VAS (I) (90 min) NPS (1 week)	NS at all times	Average VAS ↓ 72% at 90 min
Blanchard et al., 1990 (71)	n = 12; C, DB, R	+SGB, signs and symptoms (2.9 years)	Reserpine 0.5 mg UE, 1 mg LE, 30–50 ml saline, 1 block	Saline 30–50 ml (20 min)	↓ VAS (I) > 50% = relief (until no relief)	NS at all times	83% with relief at 30 min
Hord et al., 1992 (57)	n = 12; C, DB, R	+SGB, + cold stress test, history of CRPS (?)	Bretylium 1.5 mg/kg in 40–60 mg 0.5 mg L, 2 blocks	Lidocaine 0.5%, 40 ml UE, 60 ml LE, 2 blocks (20 min)	VAS (R) > 30% relief (until no relief)	Significant difference at average duration of relief with bretylium; 20 days vs. L; 3 days NS at all times	Duration of average relief with L; 3 days
Glynn et al., 1993 (72)	n = 33; C, DB, R	+Guanethidine IVRB (7 years)	Athropine 0.6 mg in 10 ml saline, 1 block	Saline 10 ml (10 min)	VAS (I) (R) (1 h) CS (1 week)		Average VAS ↓ 30% at 60 min

C, crossover; CS, categorical scale; DB, double-blind; (I), intensity; L, lower extremity; n, number of subjects in smallest group after randomization; NPS, numeric pain scale; NS, not significant; q, each; R, randomized; (R), relief; Res, reserpine; SGB, sympathetic ganglion block; UE, upper extremity; VAS, visual analogue scale for pain.

there were significant sedation and hypotension issues and the continuous epidural infusion resulted in several infections (Rauck et al., 1993). Intravenous ketanserin had no analgesic effects in a small crossover trial (Bounameaux et al., 1984).

Intravenous systemic phentolamine (an alpha-adrenergic receptor antagonist) studies have had conflicting results (Table 6). Two trials examined the phentolamine response in CRPS after an initial saline placebo infusion. One trial with a low methods score (Raja et al., 1991) found 45% of patients had significant short-term relief with phentolamine, but a much larger trial (Verdugo and Ochoa, 1994) found only 9% of patients had significant relief. Verdugo and Ochoa found that after starting the intravenous saline some patients developed a progressive placebo analgesia. They attributed the phentolamine analgesia observed in 9% of their patients, and in 45% of the patients in Raja et al.'s study to the problem of starting the phentolamine infusion during the progressive development of the placebo analgesia, which could be confused with a phentolamine effect.

An additional randomized crossover trial compared intravenous phentolamine to phenylephrine (an alpha-adrenergic receptor agonist) and demonstrated that 17% of subjects had significant relief of ongoing pain with phenylephrine and 9% had significant relief with phentolamine (Verdugo and Ochoa, 1994). These data suggest there is either no phentolamine analgesia, or an effect in only a small subset of CRPS patients (Campbell and Raja, 1995).

3.3. Comparison of methods scores

The mean methods scores were higher for the PNP trials than the CRPS trials (66.2 ± 1.5 vs. 57.6 ± 2.9 , $P < 0.01$). The CRPS trials tended to use less subjects in the smallest group than the PNP trials (18.6 ± 3 vs. 28.8 ± 3.4 , $P < 0.01$). The CRPS trials also were less likely to use placebo controls (71% used placebo vs. 94% of PNP, $P < 0.01$), perform double-blinding (57% double-blinded vs. 94% of PNP, $P < 0.001$), and to use statistical tests for differences in outcome between treatment and control groups (67% tested differences vs. 90% of PNP, $P < 0.05$). The CRPS studies utilized several study designs not seen in the PNP trials, such as a mixed crossover-parallel architecture (Blanchard et al., 1990), the use of different statistical methods or outcome measures at different post-treatment intervals (Glynn et al., 1981; Rocco et al., 1989; Glynn et al., 1993), and the use of variable lengths of follow-up depending on outcome scores (Blanchard et al., 1990; Hord et al., 1992; Jadad et al., 1995). These novel designs also reduced the methods scores.

More than one fifth of the subjects dropped out during the study in 30% of the PNP trials (vs. 13% of CRPS, $P < 0.05$). The difference between the PNP and CRPS high attrition frequencies was attributable to the side-effects of the tricyclic antidepressants, which accounted for 60% of

the high attrition PNP trials but only 30% of the total PNP trials.

The CRPS and PNP trials share several problems in their methods. No trial in either group tested for the adequacy of patient blinding, which is problematic in trials where the incidence of side-effects of treatment and placebo differ. Side-effects enhance the placebo effect by suggesting the medication has potency (Max et al., 1988a). When side-effects are unequal between groups, interpretation of the trial results is aided by asking patients which treatment they think they received and the reasons for that choice (Moscucci et al., 1987).

To prevent spurious findings of efficacy in analgesia trials which use treatments having frequent adverse effects, it has been recommended that the experimental design use a range of doses of the study drug, or compare the treatment to an active placebo which also has side-effects, or compare one active drug to another (Max et al., 1988a). This methodology was used in 43% of the CRPS trials and 42% of the PNP trials. There were seven trials that compared only active treatments, with no placebo included in the design. Four trials found one active treatment more effective than the other (Langhor et al., 1982; Watson and Evans, 1985; Watson et al., 1992; Geertzen et al., 1994). Two CRPS trials found the active treatments equally effective, but they used sympathetic ganglion blocks (SGB) with local anesthetic for the control treatment, which is problematic since there are no placebo controlled trials demonstrating SGBs are effective analgesics in CRPS (Bonelli et al., 1983; Raja et al., 1991). Another unusual design compared an alpha-adrenergic receptor antagonist to an alpha-adrenergic receptor agonist and found no analgesic differences, which demonstrated that alpha-adrenergic modulation of nociception was not observed (Verdugo and Ochoa, 1994).

Another design problem shared by the CRPS and PNP trials was the failure to evaluate long-term outcomes. The average duration of the CRPS trials was 3.5 ± 0.5 weeks and of the PNP trials 3.3 ± 0.5 weeks. In chronic pain conditions this is a problem since there is little data on long-term analgesic efficacy and on the patients ability to tolerate the adverse side-effects associated with chronic treatment.

The CRPS trials used on average only 2.7 ± 0.3 outcome measures, and the PNP trials used 2.4 ± 0.2 measures. Functional and physical measures and activities of daily living were rarely assessed and would have improved the methods scores of most studies.

4. Conclusions

The PNP data clearly indicate that tricyclic antidepressants are effective analgesics in approximately half of the patients, and these drugs have been recommended as first-line agents for all neuropathic pain except trigeminal neur-

Table 5

Controlled drug trials in CRPS

Trial (methods score)	Subjects and design	Diagnosis (duration of symptoms)	Treatment drug	Control treatment	Outcome measure (follow-up)	Results: drug vs. control	Results: control
Christensen et al., 1982 (71)	n = 10; P, SB, R	4/7 signs and symptoms (13 weeks)	Prednisone 30 mg/day up to 12 weeks	Placebo 3× day up to 12 weeks	Clinical scale (12 weeks)	Significant at 12 weeks	Clinical scale ↓ 28% at 12 weeks
Bounameaux et al., 1984 (45)	n = 9; C, DB, R	Signs and symptoms? (?)	Ketanserin i.v. bolus 10 mg	Placebo i.v. bolus	Clinical scale (25 min)	No change at 25 min	No change at 25 min
Gobelet et al., 1986 (46)	n = 10; P, O, R	Symptoms and signs of CRPS (7 weeks)	Calcitonin, s.c. 100 U/day for 3 weeks, physical therapy	Physical therapy	Clinical scales (8 weeks)	NS at all times	Significant at 3 weeks for pain, edema, ROM
Bickerstaff and Kanis, 1991 (82)	n = 20; P, DB, R	Colle's Fx, ↓ ROM, edema, vasomotor, dolorimetry (8 weeks)	Calcitonin intranasal 400 U/day for 4 weeks	Saline intranasal, 2× day for 4 weeks	Pain% and three physical measures (12 weeks)	NS at all times	Pain% and all three measures improved over 12 weeks
Gobelet et al., 1992 (80)	n = 33; P, DB, R	Kozin's criteria for definite CRPS (9 weeks)	Calcitonin intranasal 300 U/day for 3 weeks, physical therapy	Saline intranasal 3× day for 3 weeks, physical therapy	Clinical scales (8 weeks)	Significant at week 8 for pain and ROM	Significant at week 1 for pain and ROM
Glynn and Casale, 1993 (27)	n = 6; C	+G block, upper limb pain (2.1 years)	Bupivacaine 0.5% 10 ml SGB × 1	Morphine 5 mg in 10 ml saline SGB	Patient report of relief (1 h)	4/6 complete no pain relief after SGB	No relief in any patient
Rauck et al., 1993 (59)	n = 26; C, DB, R	+SGB, pain, edema, hyperalgesia (3.8 years)	Clonidine 300 µg and 700 µg epidural in 10 ml saline	Saline 10 ml epidural	VAS (1), MPQ (6 h)	Significant on VAS, MPQ from baseline for both doses	NS from baseline
Braus et al., 1994 (51)	n = 17; P, SB, R	CVA, Kozin's criteria for definite CRPS (6–8 weeks)	Methylprednisolone, 32 mg/day for 2 weeks, then taper over 2 weeks	Placebo 4× day, for 4 weeks	Clinical scale (4 weeks)	Average clinical scale ↓ 65% at week 4	No change in clinical scale at 4 weeks

C, crossover; DB, double-blind; Fx, fracture; G, guanethidine; (I), intensity; MPQ, McGill Pain Questionnaire; n, number of subjects in smallest group after randomization; NS, not significant; O, open; P, parallel; Pain%, the % of patients with residual pain; QST, questionnaire sensory testing; R, randomized; ROM, range of motion; SB, single-blind; SGB, sympathetic ganglion block; VAS, visual analogue scale for pain.

algia (Fields, 1994; Galer, 1995a). There was consistent support for intravenous and topical lidocaine, intravenous ketamine, carbamazepine and topical aspirin. There was limited support for the effectiveness of oral, topical and epidural clonidine.

The trial data was contradictory for mexiletine, phenytoin, topical capsaicin, oral non-steroidal anti-inflammatory medication and intravenous morphine. Mexiletine is probably effective, although the largest trial (Stracke et al., 1992) found it effective only in a post hoc evaluation of a subset of diabetic patients with stabbing or burning pain. Although meta-analysis of the capsaicin trials indicated an analgesic effect, all the capsaicin trials which reported an effect or trend (Bernstein et al., 1989; Chad et al., 1990; Capsaicin Study Group, 1991; Watson et al., 1993) suffer from a blinding bias due to the burning sensation associated with topical capsaicin and the use of an inert placebo. Oral non-steroidal anti-inflammatory medications are probably ineffective for PNP, with two well designed studies finding no analgesia (Max et al., 1988a; Weber et al., 1993). Intravenous morphine is probably effective in neuropathic pain, with two well designed studies demonstrating analgesia (Kupers et al., 1991; Rowbotham et al., 1991), but because of the inherent risks opioids should be used as a last resort. Codeine, propranolol, lorazepam and intravenous phentolamine have been ineffective as analgesics in various neuropathic pain states.

Despite the clinical trial data demonstrating successful pain relief with several drug regimens, the pharmacologic management of neuropathic pain is difficult. A survey of 188 specialist physicians experienced in treating neuropathic pain (Davies et al., 1991) revealed that only a minority would rate their analgesia results as good or excellent with antidepressants (40%), anticonvulsants (35%), opioids (30%) and simple analgesics (18%). This lack of success may reflect an inadequate dosage of medication, the development of drug tolerance, adverse side-effects terminating the treatment, or modest analgesic effects which do not relieve a major component of the patient's pain.

Several points emerge from the CRPS controlled trial literature. The only trial data that consistently demonstrated analgesia was with oral corticosteroids. There was limited support for the analgesic effectiveness of topical DMSO, epidural clonidine and IVRBs with bretylium and ketanserin.

The trial data was contradictory for intranasal calcitonin and intravenous phentolamine. The intranasal calcitonin studies had high methods scores and similar designs, but opposite results (Bickerstaff and Kanis, 1991; Gobelet et al., 1992). The conflicting results were probably attributable to differences in outcome measures, with the use of reliable continuous objective outcome data giving additional credence to the Bickerstaff and Kanis study which found no calcitonin effect.

The phentolamine studies also had conflicting results (Table 6). The two trials with the most patients and highest

methods scores found no phentolamine analgesia (Verdugo and Ochoa, 1994). These data suggest there is either no phentolamine analgesia, or a short-term effect in a small subset of CRPS patients (Campbell and Raja, 1995). Future phentolamine studies should have randomized crossover or parallel designs and address the issues of optimal phentolamine dosage (Raja et al., 1996), the onset time and duration of phentolamine and placebo analgesia (Fine et al., 1994; Verdugo and Ochoa, 1994; Galer, 1995b), exclusion of patients with unstable baseline pain (Raja et al., 1996), and methodology to minimize the placebo effect and to maintain double-blinded conditions (Campbell and Raja, 1995; Verdugo and Ochoa, 1995). An enriched enrollment protocol could also be utilized to focus on a subset of possible phentolamine responders (Byas-Smith et al., 1995).

There is consistent evidence that guanethidine and reserpine IVRBs are ineffective analgesics for CRPS and limited evidence that IVRBs with droperidol and atropine are ineffective. There is no placebo controlled data on SGBs with local anesthetics, surgical sympathectomy, or physical therapy.

The limited success of controlled trials using regional and systemic adrenergic blockade in CRPS is surprising in light of the frequent analgesia reported with local anesthetic sympathetic ganglion block (SGB) and the long-term pain relief observed with surgical sympathectomy. Local anesthetic SGB has been advocated as a diagnostic test to identify CRPS patients with sympathetically maintained pain, and serial SGBs have been advocated as a treatment for CRPS, with complete and lasting relief reported in 18–59% of cases (Bonica, 1979). All the SGB and surgical sympathectomy studies are outcome series without controls, except for three sympathetic block studies. One study found local anesthetic SGBs superior to morphine SGBs, but this was a small study without a randomized, double-blinded protocol, no placebo effect, no quantitative outcomes, no statistical analysis, and a low methods score (Table 5) (Glynn and Casale, 1993). Two SGB trials found no difference between SGB and 'active' controls but there is considerable evidence that the 'active' controls used in these studies (guanethidine and phentolamine) are ineffective analgesics for CRPS (Tables 3s and 6) (Bonelli et al., 1983; Raja et al., 1991). Placebo controlled trials with quantitative sensory testing are necessary to determine if SGB in CRPS patients can provide short-term analgesia without afferent fiber blockade of the plexus.

Physical therapy is widely recommended as a first line treatment for CRPS (Brody and Andary, 1993; Kozin, 1993; Subbarao and Blair, 1995; Priebe and Holmes, 1996), and has also been recommended for treatment of PNP (Fields, 1994; Galer, 1995a). Early physical therapy treatment has been advocated for CRPS, since earlier treatment correlates with better outcome (Rosen and Graham, 1957; Poplawski et al., 1983). Despite the widespread use of physical therapy in the treatment of CRPS and PNP, no controlled trials have examined its efficacy.

Table 6
Controlled phenotolamine trials in CRPS

Trial (methods score)	Subjects and design	Diagnosis (duration of symptoms)	Treatment: phenotolamine	Control treatment	Outcome measure (follow-up)	Results: phenotolamine vs. control	Results: control
Raja et al., 1991 (41)	<i>n</i> = 20; C, DB	Pain and hyperalgesia (3.1 years)	Phentolamine 25–35 mg i.v. over 15–40 min (after saline)	Saline i.v. for 1st 8–36 min (average 21 min) before drug	↓ VAS (I) > 50% = pain relief (1 h)	9/18 with relief	2/20 patients had > 80% ↓ VAS (I)
Raja et al., 1991 (48)	<i>n</i> = 20; C, O, R	Pain and hyperalgesia (3.1 years)	Phentolamine 25–35 mg i.v. over 15–40 min (after saline)	SBG with 0.25% bupivacaine 10 ml UE, 20 ml LE	↓ VAS (I) > 50% = pain relief (1 h)	NS	9/18 with relief after SGB
Verdugo and Ochoa, 1994 (52)	<i>n</i> = 76; C, SB	IASP 1986 criteria for CRPS, with ongoing pain (?)	Phentolamine 35 mg i.v. over 30 min (after saline)	Saline i.v. for 1st 30 min before drug and 30 min after drug	NPS ongoing and evoked (allodynia and pressure) ↓ NPS > 50% = pain relief (1 h)	NS, 9% relief ongoing pain, 3% relief of allodynia, and 6% relief pressure pain	29% relief of ongoing pain, 28% allodynia, and 18% relief of pressure pain
Verdugo and Ochoa, 1994 (62)	<i>n</i> = 23; C, DB, R	IASP 1986 criteria for CRPS, with ongoing pain (?)	Phentolamine 35 mg i.v. over 20 min, saline i.v. 20 min at start and end of protocol	Phenylphrine 500 µg i.v. over 20 min, either before or after phentolamine	NPS ongoing and evoked (allodynia and pressure) ↓ NPS > 50% = pain relief (1 h)	NS, 9% relief ongoing pain 11% relief of allodynia, and 15% relief pressure pain	17% relief of ongoing pain, 0% relief of allodynia, 8% relief of pressure pain

C, crossover; DB, double-blind; (I), intensity; *n*, number of subjects in smallest group after randomization; NPS, numeric pain scale; NS, not significant; O, open; R, randomized; SB, single-blind; SGB, sympathetic ganglion block; VAS, visual analog scale for pain.

The control groups in two large calcitonin trials used early CRPS patients (8 weeks after trauma), and these control groups had a similar reduction in pain during the 8-week trial period. One trial gave its control patients 8 weeks of active and passive physical therapy within the pain free range of movement, positive pressure treatments for edema, lymphatic drainage or whirlpool, TENS and cooling (Gobelet et al., 1992), while the other trial gave its controls no treatment (Bickerstaff and Kanis, 1991). Placebo intranasal saline treatments were given for 3–4 weeks to both control groups, and after 8 weeks the reduction in pain was virtually identical (55–57%) for the physical therapy and no treatment control groups. This comparison illustrates the need for random-ized trials to distinguish between the spontaneous resolution of pain in early CRPS and possible effects of physical therapy.

The mexiletine, capsaicin, oral non-steroidal anti-inflammatory medication, intravenous morphine, and calcitonin treatment trials were all considered for meta-analysis. Each treatment had three or more trials with contradictory results, but only the capsaicin trials presented adequate homogeneous data to perform a meta-analysis. Most PNP and CRPS trials used small numbers of subjects and pooling the results would help compensate for variability in the data. The use of meta-analytic methods should be a consideration in trial design. Future trials need to indicate the number of subjects with a positive response in the treatment and control groups at each evaluation interval, and to replicate the methods of earlier investigations so it would be reasonable to compare the outcomes.

The mean methods scores were higher for the PNP trials than the CRPS trials. The CRPS trials tended to use less subjects, were less likely to use placebo controls, were less likely to blind the patients and evaluators, and less likely perform statistical tests for differences in outcome measures between groups. The PNP trials had more frequent patient attrition than the CRPS trials due to the side-effects of the tricyclic antidepressants. No trial in either group tested for the adequacy of patient blinding and the trial outcome measures usually failed to evaluate function, physiology, or activities of daily living.

There were only nine long-term treatment trials, which looked at six different drugs. Further trials are needed to establish long-term analgesic efficacy for all treatments currently used in chronic neuropathic pain and CRPS. The two PNP trials that examined analgesic efficacy for 3 months (or longer) demonstrated no relief with phenytoin (Saudek et al., 1977) or capsaicin (Low et al., 1995). Seven CRPS studies have followed patients for 3 months (or longer). Only prednisone provided effective analgesia for 3 months when compared to placebo (Christensen et al., 1982). Three studies found no outcome differences between guanethidine IVRBs compared to placebo (Blanchard et al., 1990), SGBs (Bonelli et al., 1983), or multiple guanethidine IVRB treatments (Ramamurthy et al., 1995). One study found no analgesic effect for repeated guanethidine IVRBs compared

to physical therapy, except for a reduction of dolorimetry scores developing 5 months after the completion of guanethidine treatments (Field and Atkins, 1993). Reserpine IVRBs (Blanchard et al., 1990) and intranasal calcitonin (Bickerstaff and Kanis, 1991) were also ineffective long-term analgesics in CRPS.

The patients in the PNP trials were not systematically evaluated for symptoms or signs of abnormal sudomotor or vasomotor function, which if present would also classify the patients as having CRPS II. The practice of separating chronic pain states deriving from nerve injuries into PNP and CRPS II diagnostic categories would be of clinical significance if there are differences in the treatment response or natural history of the conditions.

Anecdotal evidence suggests that effective analgesics for neuropathic pain may also help some CRPS patients, but controlled drug trials are needed to establish such a role. Surprisingly, the only overlaps in the treatments used for PNP and CRPS were intravenous phentolamine and epidural clonidine. Intravenous phentolamine was found ineffective in PNP (Verdugo et al., 1994) and in most cases of CRPS (Raja et al., 1991; Verdugo and Ochoa, 1994). Epidural clonidine was an effective analgesic in PNP (Eisenach et al., 1995) and CRPS (Rauck et al., 1993). The similar treatment outcomes for PNP and CRPS patients do not support the clinical utility of using separate diagnostic categories.

Can separating patients into CRPS I and II predict treatment results? The only trial that separately evaluated outcomes for CRPS I and II patients was an intravenous phentolamine trial, which found no analgesia in either diagnostic category (Verdugo and Ochoa, 1994).

Another question raised by the controlled trial data is whether the type of neuropathy or etiology of the CRPS has any correlation to the treatment response. There are no CRPS trial data which looked at etiologies for CRPS and treatment outcomes. The PNP trials demonstrated similar effectiveness for antidepressants and intravenous lidocaine in diabetic, postherpetic, and peripheral nerve injury PNP. Mexiletine was effective in diabetic and peripheral nerve injury PNP, and intravenous morphine and ketamine were both effective for postherpetic and peripheral nerve injury PNP (Table 1).

The PNP trial literature gave consistent support for analgesic effectiveness of tricyclic antidepressants, intravenous and topical lidocaine, intravenous ketamine, carbamazepine and topical aspirin. The CRPS trial data gave consistent support for corticosteroid analgesia and consistently found that IVRBs with guanethidine or reserpine were ineffective. Long-term pain relief was demonstrated in only one CRPS corticosteroid trial. The CRPS trials had lower average methods scores than the PNP trials, and tended to use less subjects and were less likely to use a placebo, double-blinding, or perform statistical data analysis. There was almost no overlap in the controlled trial literature between treatments for PNP and CRPS, and treatments used in both conditions had similar results.

PNP and CRPS studies accounted for 0.2% and 0.1%, respectively, of the 8243 controlled pain studies published between 1950 and 1990 (Jadad et al., 1996). Clearly the paucity of controlled trials for PNP and CRPS underlies the difficulties frequently encountered by the clinician in the treatment of these chronic pain states (Davies et al., 1991). The publication rate for controlled pain studies is rapidly accelerating and over half of all PNP and CRPS trials have been published since 1990. Well designed randomized controlled trials will provide the future foundation for the successful management of PNP and CRPS.

5. Addendum

An additional controlled treatment study for neuropathic pain came to the author's attention too late for inclusion in this review (Cohen et al., 1990). This study compared both clonidine and pentoxifylline to placebo in diabetic neuropathic pain patients and found no analgesic effects for either drug.

Appendix A: Scoring criteria for evaluation of methods in controlled clinical trials for peripheral neuropathic and complex regional pain syndromes

Criterion	Weight
<i>Study population</i>	
A. Inclusion/exclusion criteria listed	2
B. Randomized	6
C. Number of subjects in smallest group after randomization (three points for each 10 subjects, up to 50 subjects)	15
D. No baseline differences between groups for:	
Duration of symptoms	2
Outcome measures	8
E. Patient attrition at last outcome measure:	
<20%	2
<10%	2
<i>Interventions</i>	
F. Treatment protocol explicitly described	3
G. Comparison with placebo treatment	8
H. Comparison with another treatment, dose, or placebo with side-effects	5
<i>Effect</i>	
I. Blinded patients:	
Attempted blinding	4
Blinding evaluated and successful	4
J. Blinded outcome assessment:	4
K. Outcome measures: two points for each measure, maximum score of 10	10
L. Longest follow-up period comparing outcome measures between groups (one point for each week after onset of treatment, up to 13 weeks)	13
<i>Data presentation and analysis</i>	
M. Statistical tests for effect differences between treatment groups ($P < 0.05$)	12
Maximum possible score	100

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